Multi-Expert Agentic Pipeline for Peptide Generation

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Abstract

We present a proof-of-concept multi-expert agentic pipeline for *in silico* peptide generation [7]. A central orchestrator coordinates three specialized agents: an AI Designer, a Chemistry Constraints agent, and a Biology agent providing an embedding-based ranking signal [8]. The generator is based on variational architectures (VAE/CVAE), and chemistry and biology scores are fused by the orchestrator to build a global Top-K ranking across iterations.

Keywords: Multi-Agent Systems, Peptide Generation, In Silico Drug Design, VAE/CVAE, Evolutionary Scale Modeling.

1 Introduction

Peptides, and especially antimicrobial peptides (AMPs), are promising candidates for therapeutic, agricultural, and industrial applications [7]. Their *in silico* design is typically formulated as an iterative optimization process: generate candidate sequences, apply formal constraints, and evaluate each candidate with proxy scores for activity, toxicity risk, and stability [7, 8]. While recent machine learning methods have accelerated sequence design, balancing multiple and sometimes conflicting objectives remains difficult in practice.

This project addresses that challenge through an agentic proof of concept in which specialized software agents cooperate in a closed optimization loop. The pipeline includes an AI Designer agent that proposes candidates, a Chemistry Constraints agent that filters feasibility using proxy physicochemical rules, and a Biology agent that provides an embedding-based proxy for biological relevance. An orchestrator manages the iterative cycle (propose → filter → score → revise), applies stopping criteria, and stores logs for traceability.

The work is centered on AI aspects (modeling, orchestration, reasoning, and quantitative evaluation) and assumes no prior expertise in wet-lab biology or chemistry. Evaluation is explicit and reproducible: we measure validity, average and Top-K score, diversity, pass rates for constraints, and score evolution across rounds, then compare against a non-agentic baseline. This framing allows us to test whether multi-expert coordination provides measurable gains over single-pass generation .

The remainder of this paper is structured as follows: Section 2 provides a brief overview of the project context and related work. Section 3 details the architecture of our multi-agent pipeline and the specific roles of the AI, Chemistry, and Biology agents. Section 4 outlines our experimental setup and quantitative evaluation metrics. Section 5 presents our findings, including baseline comparisons and ablation studies. Finally, Section 7 concludes the report and discusses potential future work.

2 Context and Related Work

2.1 Evolution of In Silico Peptide Generation

The computational identification and design of therapeutic peptides have evolved from rule-based filtering to data-driven learning pipelines [1]. Early *in silico* methods relied on handcrafted descriptors and QSAR-style modeling, using aggregate physicochemical indicators (e.g., charge, hydrophobicity, or sequence length) as proxies for feasibility and activity [1]. These descriptors are still widely used as practical constraints, but current workflows increasingly combine them with learned sequence representations.

In recent years, generative and predictive models have enabled large-scale exploration of peptide sequence space. Broadly, modern systems map discrete sequences into continuous representations, generate candidate peptides under

configurable objectives, and score them with learned or heuristic predictors. This shift has improved throughput and candidate quality, but also highlighted a persistent challenge: no single model reliably satisfies multiple objectives at once (activity, safety, synthesizability, stability), which motivates more structured optimization workflows.

2.2 The Shift to Agentic Workflows

Despite these advances, peptide design pipelines often remain fragmented: generation, constraint checking, and biological scoring are implemented as separate components with limited coordination [6]. As a result, one-shot generation frequently produces candidates that satisfy part of the objective but violate other key constraints.

Agentic workflows address this by organizing the process as an explicit iterative loop: propose candidates, evaluate them with specialized modules, revise, and repeat [3]. Rather than relying on a single monolithic predictor, this design decomposes the objective into complementary roles and makes trade-offs explicit at each iteration. For constrained peptide optimization, the essential benefit is the coordination mechanism itself: specialized agents, shared evaluation signals, and closed-loop refinement [1].

2.3 Project Scope and Alignment

This project is scoped as an AI systems Proof of Concept for iterative peptide design. Concretely, the expected deliverable is an end-to-end multi-expert pipeline with three specialized agents and one orchestrator, where each iteration performs candidate proposal, chemistry-oriented constraint filtering, biology-oriented critique, and revision. The scientific objective is not to claim new biological mechanisms, but to test whether explicit role decomposition and orchestration improve optimization behavior under practical proxy constraints.

The project was supervised by an interdisciplinary team (AI, chemistry, and biology), which helped keep the work aligned with domain expectations, informed the definition of proxy constraints and evaluation criteria, and supported critical review of the reported results.

The work remains strictly *in silico*: no wet-lab validation is assumed, and all scores are documented proxies. Alignment with the project brief is ensured through quantitative reporting on validity, average and Top-K quality, diversity, constraint pass rates, and score evolution across rounds, followed by comparison to at least one non-agentic baseline. In this framing, success is defined by measurable and reproducible gains from iterative coordination rather than by absolute biological performance claims.

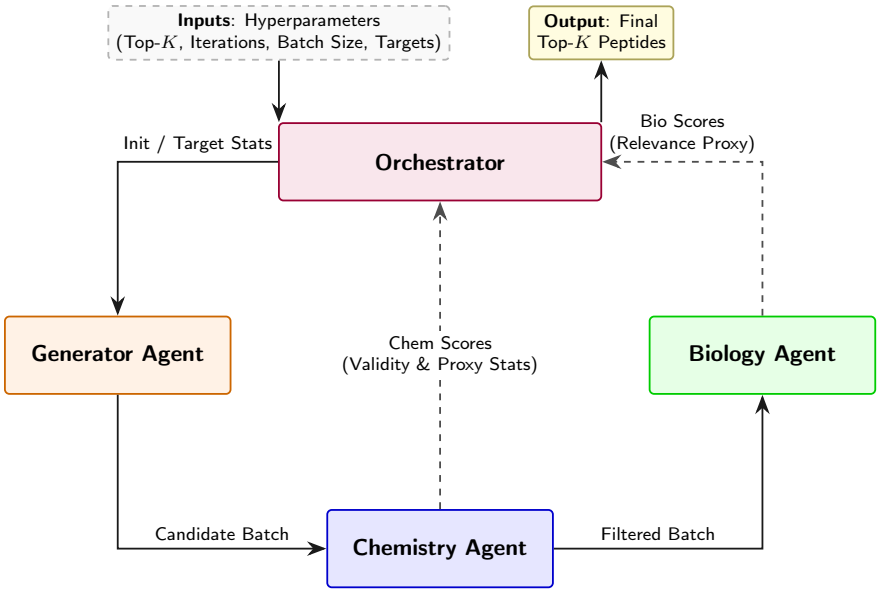

Figure 1: Agentic Pipeline Architecture. Solid lines denote the sequential flow of peptide candidate batches. Dashed lines indicate the feedback loops, where both the Chemistry and Biology agents return their respective proxy scores back to the Orchestrator. Color coding differentiates the specialized domains of the pipeline.

3 Multi-Agent System Architecture

The global pipeline is organized around three specialized agents coordinated by a central orchestrator. Figure 1 summarizes the end-to-end data flow and feedback loops in the iterative process.

3.1 The Orchestrator

The Orchestrator is the control layer that turns the three experts into a single iterative optimization system. Its role is to enforce a fixed sequence of operations:

1. Generate a batch of candidates from target constraints (user-provided or inferred from the chemistry configuration).
2. Pass this batch through the Chemistry Agent to compute physicochemical descriptors and in-limits status; if no candidate is in-limits, the full chemist output is still forwarded for biological scoring to avoid stalling the loop.
3. Based on these expert outputs, the Orchestrator builds a global ranking of unique peptides across rounds, keeps the Top-K output at the end, and uses top-ranked candidates as parents for the next generation cycle.

In this way, each iteration combines exploration (new proposals) and exploitation (retaining best current candidates) under one consistent decision policy. Stopping criteria are handled at the orchestration level rather than inside each expert module. In practice, the loop is driven by a fixed iteration budget; when

an intermediate stage fails for a given round (for example no usable candidates), the run continues to the next round:

- A maximum number of iterations is reached
- Some rounds may contribute no candidates to the global ranking because generation or filtering yields no usable sequences

This implementation supports reproducible experimentations: changing run behavior only requires updating orchestrator-level parameters, with no need to modify the expert modules. Compared to the previous global behavior, two points are now explicit: the process keeps a global best-by-sequence pool across all rounds (instead of only round-local selection), and chemistry out-of-limits candidates can still be biologically scored when no in-limits candidates exist. This agent also provides traceability by recording round-wise pipeline events and outcomes:

- Input settings
- Filtered populations
- Expert scores
- Selected candidates

This log-centric design makes the optimization path auditable from end to end: one can reconstruct why a peptide was retained or discarded at a given step, compare runs under different settings, and support post-hoc analysis of convergence behavior and failure cases.

3.2 Generator Agent

To generate novel peptide candidates, two generative architectures were implemented: a Variational Autoencoder (VAE) and a Conditional Variational Autoencoder (CVAE). Both models operate on peptide sequences encoded as fixed-length one-hot vectors, where each sequence position is represented over the 20 canonical amino acids. For a peptide of length L, the input dimensionality is therefore $20 \times L$. The objective of these models is to learn a structured latent representation of peptide sequences that can be sampled in order to generate new candidates. The VAE provides unconstrained generation based on the global training distribution, whereas the CVAE extends this framework by conditioning generation on physicochemical descriptors, enabling more directed design.

Variational Autoencoder

The VAE is a probabilistic generative model that learns to encode peptide sequences into a continuous latent space and decode latent samples back into

amino acid sequences. Given an input peptide representation x, the encoder predicts the parameters of a Gaussian latent distribution:

$$q_\phi(z \mid x) = \mathcal{N}(\mu_\phi(x), \sigma_\phi^2(x)). \tag{1}$$

A latent vector z is then sampled using the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \tag{2}$$

which allows gradients to propagate through the sampling step during training.

In the implemented architecture, the encoder is a multilayer perceptron composed of linear layers, layer normalization, and ReLU activations. The output is split into two vectors corresponding to the latent mean and log-variance. The decoder is another multilayer perceptron of symmetric structure that maps the latent vector back to the original sequence space.

Rather than reconstructing the sequence using a mean squared error loss, the decoder outputs raw logits for each amino acid at each sequence position. These logits are reshaped into categorical distributions over the 20 amino acids, and the reconstruction loss is computed using cross-entropy. This choice is more appropriate for peptide sequences, since amino acids are discrete categories rather than continuous values.

The training objective combines reconstruction quality and latent regularization:

$$\mathcal{L}(x) = \mathcal{L}_{\text{recon}}(x, \hat{x}) + \beta\, D_{\text{KL}}\big(q_\phi(z \mid x) \,\|\, p(z)\big), \tag{3}$$

where $p(z) = \mathcal{N}(0, I)$ is the prior distribution and D_{KL} denotes the Kullback–Leibler divergence. In practice, the KL term is progressively increased during training through KL annealing in order to stabilize optimization and reduce posterior collapse at early epochs.

The model is optimized with Adam, combined with cosine annealing of the learning rate and gradient clipping to improve training stability. During generation, latent vectors are sampled from the standard normal distribution and decoded into amino acid probability distributions. Sequence diversity is controlled through a temperature parameter applied before softmax sampling.

Conditional Variational Autoencoder

The CVAE extends the VAE by conditioning both encoding and decoding on an auxiliary vector y describing target peptide properties. Instead of modeling only the unconditional peptide distribution, the CVAE learns a conditional distribution:

$$q_\phi(z \mid x, y), \qquad p_\theta(x \mid z, y). \tag{4}$$

In the implemented model, the condition vector is concatenated with the peptide input before the encoder and concatenated again with the latent representation before the decoder. This architecture allows the model to generate peptides not only consistent with the training distribution, but also biased toward a specified physicochemical profile.

The condition vector is constructed from a fixed ordered list of physicochemical descriptors:

- length,
- molecular weight,
- net charge at pH 5.5,
- isoelectric point,
- hydrophobicity,
- cathionicity,
- hydrophobic moment,
- logP,
- Boman index,
- number of hydrogen bond donors,
- number of hydrogen bond acceptors,
- topological polar surface area (TPSA).

In the current implementation, each property is encoded as a scalar value. Missing entries are replaced by predefined defaults, user-provided constraints override these defaults, and the resulting vector is padded or truncated to match the configured conditioning dimensionality.

This conditioning strategy enables the user to steer generation toward peptides with desired physicochemical characteristics. For example, the model can be guided toward sequences with a target range of charge, hydrophobicity, or molecular weight. In that sense, the CVAE is better suited than the VAE for directed peptide design.

Comparison Between VAE and CVAE

The VAE and CVAE serve related but distinct purposes. The VAE is an unconditional model: it learns the overall distribution of the training peptides and can sample novel sequences from this learned space. It is therefore useful for exploratory generation and for assessing whether the latent space captures relevant structural patterns in peptide sequences.

By contrast, the CVAE introduces control over the generation process. Because the model is explicitly conditioned on target descriptors, it can be used to bias sampling toward peptides exhibiting specific physicochemical properties. This makes the CVAE more appropriate for inverse design settings, where the goal is not merely to generate plausible peptides, but to generate candidates with a desired profile.

In summary, the VAE emphasizes diversity and distribution learning, whereas the CVAE emphasizes controllability and property-guided generation.

Physicochemical Conditioning

The CVAE conditioning space includes 12 descriptors, while the current Chemistry Agent actively evaluates six configurable properties (length, molecular weight, net charge, isoelectric point, hydrophobicity, and logP). This means that generation can be guided by a broader condition vector than the one strictly validated by chemistry filtering.

This link between chemistry-oriented descriptors and the generative model provides a practical way to translate biochemical objectives into numerical conditioning variables. As a result, the CVAE can be integrated into a broader peptide design pipeline in which candidate peptides are generated, analyzed, and filtered according to downstream criteria.

Limitations

Several limitations must be acknowledged. First, the current implementation assumes fixed-length peptide inputs, since the one-hot encoding and decoder output dimensionality depend directly on sequence length. As a result, the model does not naturally support variable-length peptides without additional preprocessing or architectural changes.

Second, the conditioning mechanism in the CVAE should be interpreted as a soft constraint rather than a hard constraint. The condition vector influences the probability distribution learned by the decoder, but it does not guarantee that every generated peptide will strictly satisfy the requested bounds. Consequently, an additional post-generation filtering step may still be necessary to enforce physicochemical thresholds.

Third, the usefulness of the CVAE depends strongly on the quality and consistency of the condition labels used during training. If the model is trained on sequences that are not paired with meaningful physicochemical targets, the conditioning mechanism will not learn a reliable relationship between conditions and generated sequences.

Finally, neither the VAE nor the CVAE directly predicts antimicrobial efficacy or biological activity. These models generate plausible peptide candidates from a learned sequence distribution, but downstream validation remains necessary to assess activity, selectivity, toxicity, and synthetic feasibility.

3.3 Chemistry Agent

The Chemistry Agent is the deterministic constraint-and-scoring stage of the pipeline. At a global level, it receives peptide sequences from the generator, evaluates them against user-defined physicochemical targets, and returns both a feasibility decision and a quantitative score used downstream by the orchestrator. The behavior of this agent is driven by a configurable target profile (ranges, target values, optional weights, and pH), which allows the same workflow to be reused under different experimental objectives.

Its workflow can be summarized through three complementary parts:

1. Sequence validation and property extraction: The agent first validates peptide strings against the 20 standard amino acids. For each valid candidate, it computes configured physicochemical properties using peptide-level functions and RDKit-based logP estimation.

2. Multidimensional property profiling: To characterize developability, the agent computes a configurable set of descriptors from two complementary sources: peptide-centric properties (length, net charge at chosen pH, isoelectric point, hydrophobicity) and molecule-centric properties (notably RDKit-based $\log P$). This profile lets the orchestrator compare candidates in a unified physicochemical space.

3. Quantitative scoring and filtering: The agent assigns each candidate a continuous score based on distance to target values, with optional per-property weighting. Chemistry evaluation first computes per-peptide properties and in-limit status, then returns scored candidates to the orchestrator. At orchestration level, ranking prioritizes in-limit candidates; if none are available in a round, out-of-limit candidates are still forwarded for biological scoring to preserve loop continuity. This design prevents early pipeline collapse while maintaining selection pressure toward target profiles.

The Chemistry Agent is called immediately after generation and provides the first deterministic screening stage before biological scoring. Based on user-defined limits, it sends scores and computed properties back to the orchestrator. At a higher level, this output acts as a consistent proxy signal that complements the biological score during candidate ranking.

3.4 Biology Agent (Function & Risk)

The Biology Agent evaluates the generated candidates and provides the biological signal used by the iterative loop. In our implementation, it is based on Meta/Facebook's Evolutionary Scale Modeling family (ESM-2) [2,9] and can be conditioned by context (for example an existing peptide reference or requested target properties).

At each iteration, the Biology Agent scores the current valid batch and returns values used by the orchestrator to rank candidates and keep the Top-K peptides for the next round. This makes it the core selection component on the biological side of the pipeline.

As a design choice, this score is treated as a relative proxy to guide search, not as a direct experimental measurement. The detailed scoring formulation is reported in Section 4.1.

4 Experimental Setup

This section documents the implemented scoring and comparison protocol used in the current proof of concept.

4.1 Quantitative Scoring System

Biologist Agent Scoring

The objective is to rank a batch of peptides and identify which candidate is closest to the provided context (an existing peptide or requested properties represented as a reference embedding).

The biologist module uses ESM-2 embeddings (default: `facebook/esm2\_t12\_35M\_UR50D`) with attention-mask-aware mean pooling to produce one vector per peptide.

Two proxy variants are implemented. The first uses cosine similarity, mapped to $[0, 1]$. The second (Global L2) computes Euclidean distance to a reference embedding and applies a batch-independent exponential mapping. In preliminary tests, cosine scores showed limited contrast for short peptides, so the Global L2 variant was preferred for iterative ranking.

The resulting score is used by the orchestrator for candidate ranking and Top-K selection. It is treated as a proxy signal to guide search, not as an absolute biological activity measurement.

Chemist Agent Scoring

To prevent pipeline starvation (where strict filters might discard the entire population early in the optimization process), the orchestrator applies a simple fallback policy: if no in-limits candidate is available after chemistry evaluation, biological scoring is computed on the full chemist output (including out-of-limits candidates).

The chemistry workflow therefore operates in two practical steps:

1. **Hard constraint check and scoring:** Candidates are evaluated against strict limits and scored from target-distance-based property scores.

2. **Fallback at orchestration level:** If no candidate satisfies all limits, out-of-limits candidates are still passed to biology scoring to keep the iterative loop active.

This design ensures a continuous flow of candidates to the Biology Agent while maintaining selection pressure toward developable molecules.

- **Solubility & Stability proxies:** Isoelectric point and net charge are computed for each peptide.

- **Permeability proxy:** The profile includes $\log P$ (RDKit-based) and hydrophobicity.

- **Basic developability descriptors:** Length and molecular weight are included in scoring and admissibility checks.

Orchestrator-Level Scoring and Ranking

At orchestration level, each candidate receives a combined proxy score obtained by averaging Chemistry and Biology signals with equal weight:

$$S_{\text{combined}} = \frac{S_{\text{chem}} + S_{\text{bio}}}{2}. \tag{5}$$

The ranking policy then prioritizes chemically admissible peptides (in-limits) and, within that constraint status, orders candidates by combined score (with expert scores used as tie-breakers). The orchestrator maintains this ranking globally across rounds, retaining one best entry per unique sequence and returning the final Top-K from this global pool.

In practical terms, this means candidate quality is not decided only round by round: a strong peptide discovered early can remain competitive at the end if later rounds do not surpass it.

4.2 Evaluation Metrics

We report six core metrics, computed at run level and on final Top-K outputs:

- **Validity rate (%):** fraction of returned peptides that satisfy all configured chemistry constraints (`in_limits=True`).
- **Top-1 score:** best combined score among returned candidates.
- **Top-K mean score:** average combined score across final Top-K peptides.
- **Constraint pass rate:** proportion of generated peptides passing chemistry constraints before final ranking.
- **Diversity (sequence-level):** mean pairwise distance (normalized edit distance) and unique-ratio in Top-K.
- **Diversity (distribution-level):** bigram entropy over amino-acid pairs as a complementary compositional diversity proxy.

At each iteration, chemistry and biology signals are merged through the orchestrator-level score:

$$S_{\text{combined}} = \frac{S_{\text{chem}} + S_{\text{bio}}}{2}. \tag{6}$$

The final ranking is built globally across rounds (best-by-sequence retention), then truncated to Top-K.

4.3 Baselines

The baseline is designed as a simpler non-agentic reference to test whether the proposed multi-expert iterative pipeline provides measurable gains. Concretely, we use a standalone Conditional Variational Autoencoder (CVAE) that directly

generates peptide candidates from conditioning features, without chemistry/bio critique loops and without orchestrated revision rounds. This baseline CVAE is trained on the dataset built with `database/get_data.py`. The extraction script queries DBAASP asynchronously and retains sequence-level and physicochemical fields used for conditioning (length, pH, molecular weight, logP, net charge, isoelectric point, hydrophobicity, and cationicity).

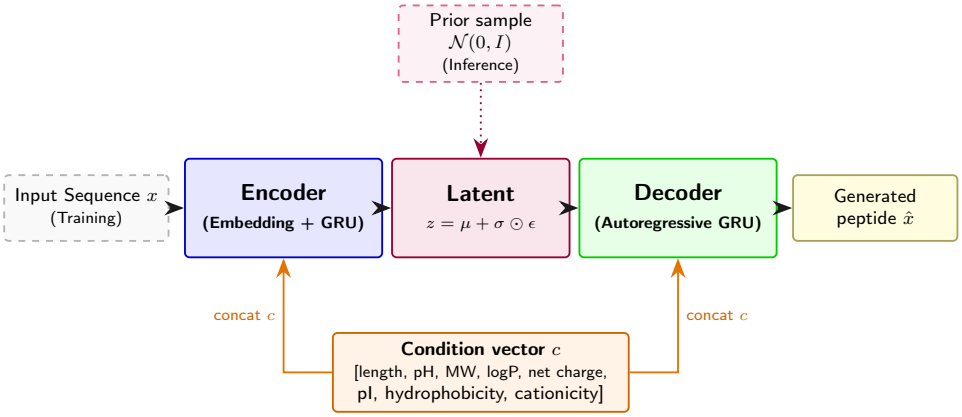

Figure 2: Baseline CVAE architecture used for comparison. During training, the model learns $q_\phi(z \mid x, c)$ and reconstructs sequences with an autoregressive decoder conditioned on z and c. During inference, peptides are generated by sampling $z \sim \mathcal{N}(0, I)$ and decoding once, without iterative multi-agent feedback.

In practice, many API entries contain missing values for one or more properties. The baseline data pipeline therefore substitutes absent fields with fixed defaults before standardization. This dataset is thus not optimal from a data-quality perspective, but for a proof-of-concept objective we considered it sufficient to establish a consistent comparison point. From an implementation perspective, the baseline model is a GRU-based CVAE. The encoder embeds amino-acid tokens and maps sequence-condition pairs to latent Gaussian parameters; the decoder is autoregressive and conditioned on both latent code and property vector. Training uses cross-entropy reconstruction loss with KL regularization, Adam optimization, and a progressive KL weight (annealed β). At inference time, candidate peptides are sampled from the latent prior under a target condition vector (scaled with the training `StandardScaler`), with temperature and top-k sampling to control exploration. This baseline serves only as the comparison anchor: unlike the full pipeline, it performs one-shot conditional generation and does not benefit from multi-agent feedback, constraint-aware iteration, or round-level refinement.

5 Results and Analysis

This section focuses on the three key curves used to understand pipeline behavior: the effect of iterations, the number of generated peptides, and target peptide length. These are not the only analyses performed, but they are the most informative under the page constraints of this report.

5.1 Effect of Number of Iterations

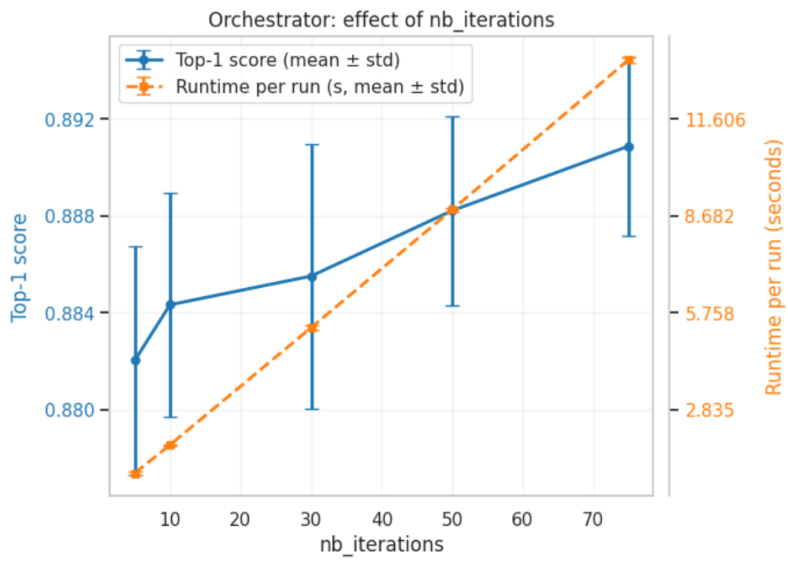

Figure 3: Effect of `nb_iterations` on Top-1 score and runtime.

Increasing iterations improves quality, but the gain remains gradual: Top-1 score moves from about 0.88 to 0.89. Runtime, however, grows sharply (from about 0.92s to 13.39s). This reflects a clear quality–time trade-off, with diminishing score gains at high iteration counts.

5.2 Effect of Number of Generated Peptides

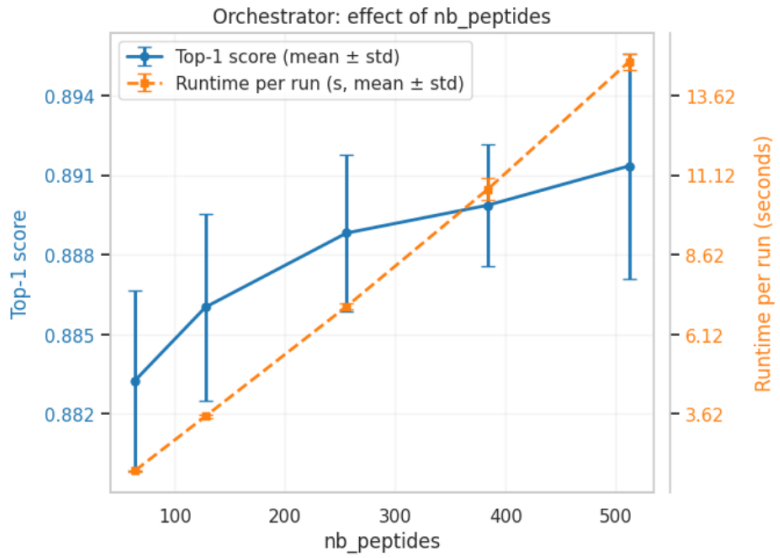

Figure 4: Effect of `nb_peptides` on Top-1 score and runtime.

When more candidates are generated per iteration, the score improves steadily (around 0.88 to 0.89), while runtime increases from about 1.83s to 14.70s. This confirms the same pattern: broader search helps, but computational cost rises quickly. A range of 256 to 384 peptides per iteration appears to be a practical compromise.

5.3 Effect of Target Peptide Length

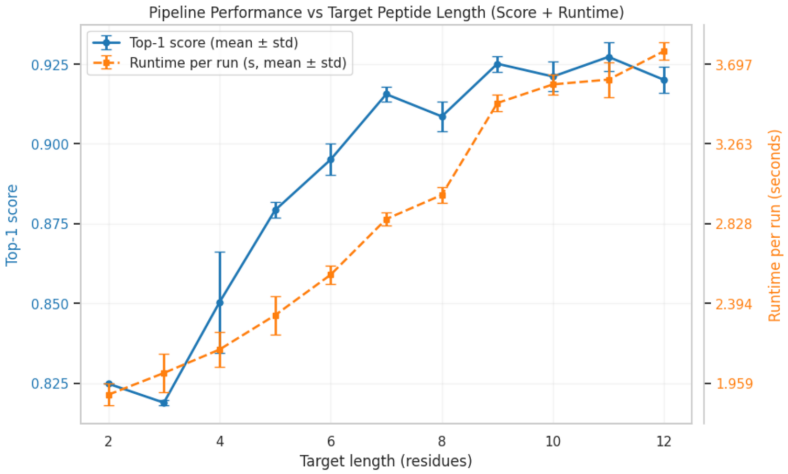

Figure 5: Top-1 score and runtime versus target peptide length.

Target length has a strong impact on both quality and speed. Very short targets (2-3 residues) are the most difficult and stay around 0.82 score. Performance then rises and stabilizes near 0.92 for longer targets. Runtime increases more smoothly, from about 1.90s to 3.77s as length grows.

Overall, the three plots tell a consistent story: better scores are achievable with larger search budgets, but the gain is moderate compared to the runtime increase. This is why selecting balanced settings is important for practical use.

6 Discussion & Future Work

6.1 Chemist agent improvement

In the current implementation, chemistry filtering and scoring are computed directly from peptide sequences. An extension could incorporate richer molecular representations (for example SMILES-derived descriptors) as additional conditioning and scoring signals. Such an extension would require a dedicated preprocessing pipeline, a clear descriptor-selection strategy, and retraining of the generation components to ensure consistency between conditioning variables and chemistry constraints.

6.2 LLM usage for input

Initially, we considered using a large language model to guide peptide selection by training it to understand and communicate preferences about desired

peptide characteristics. This approach could have provided an intuitive interface for specifying optimization targets and constraints. This LLM-based input method remains a promising avenue for future work, particularly for enhancing the usability and flexibility of the system when dealing with complex or evolving biological requirements.

6.3 Future Work: Peptide-Specific Biological Scoring

The main improvement is to replace the current proxy with a model trained (or fine-tuned) specifically on peptides. Ideally, this model should score or classify a peptide conditionally on a target (or target-related context), instead of relying on a general protein-space similarity signal.

The expected benefit is a more accurate and reliable biological signal for ranking candidates in the loop. A peptide-specialized model could also provide richer outputs than a single proxy score, for example clearer estimates of expected activity and additional information about risk or toxicity.

In short, our current biologist agent is a practical baseline for iterative optimization, while a target-conditioned peptide model is the natural next step to improve biological relevance and decision quality.

7 Conclusion

TODO: Summarize that the multi-expert agentic pipeline successfully demonstrates iterative improvement in peptide design. TODO: Mention that all code, scripts, and logs are provided in the Git repository[cite: 32, 33].

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